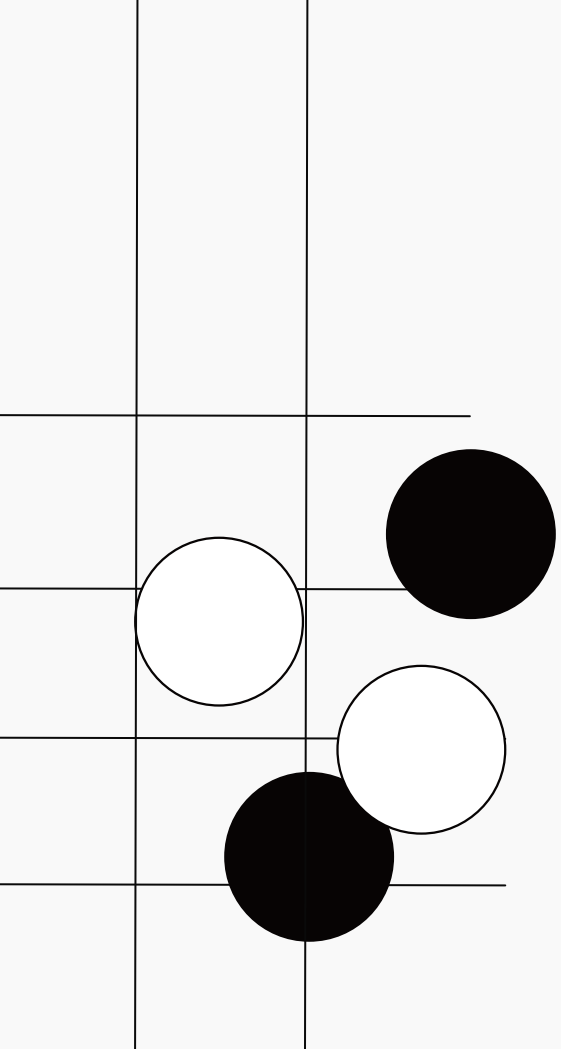




Presentation on

ZetaGo: Supervised Move Prediction on 7x7 Go

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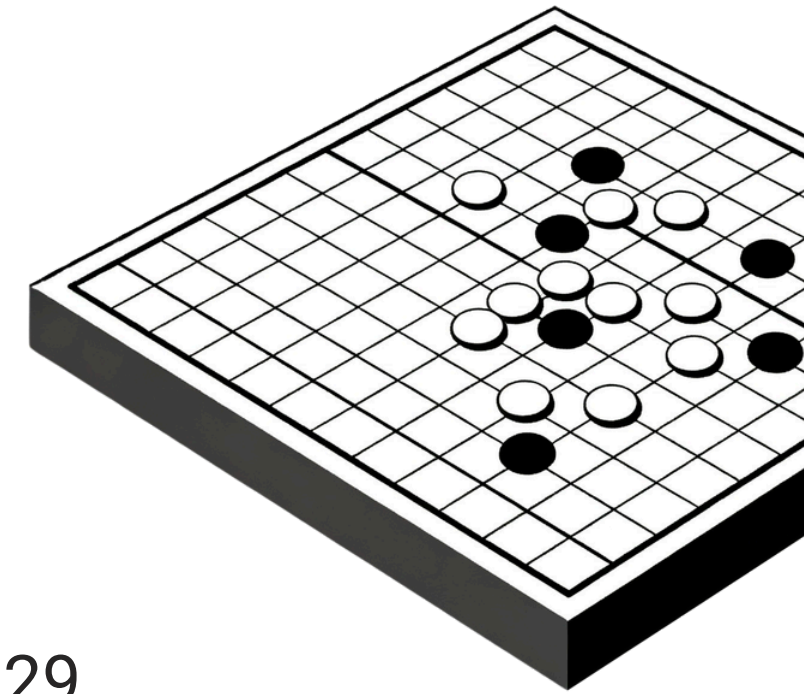
Course : CSE 4622

Date : 04.08.2026

Big Picture & Motivation

The Challenge of Computer Go

- **Go Complexity:** Has a massive state space of positions, causing classical search trees to fail.
- **AlphaGo Breakthroughs:**
 1. AlphaGo (2016) used human games + RL and it has 48 features
 2. AlphaGo Zero (2017) proved self-play surpasses human ceilings with 17 features that only contain the last 8 board states.
- **The Compute Barrier:** AlphaGo Zero required : 40 days on 4 TPUs across 29 million games, making ablation studies inaccessible to standard academic labs.



Research Question

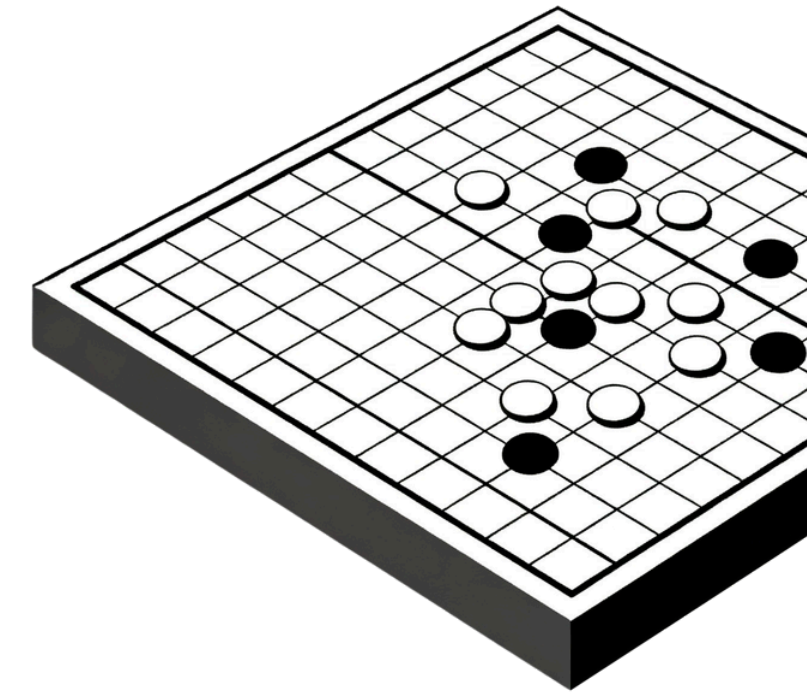
Core Question & ZetaGo Strategy

- **Core Research Question:**

As labeled self-play training data grows, do hand-crafted feature planes still help move prediction for every model, or only capacity-limited ones?

7x7 Controlled Lab: Keeps game dynamics non-trivial while enabling a full 300-cell sweep (5 models x 3 encodings x 4 dataset sizes x 5 seeds) in < 12 hours on a single GPU.

Supervised Oracle Setup: Uses KataGo (v1.16.5) as a deterministic teacher to isolate feature reliance and model capacity from reinforcement learning noise.



Related Work & Key Novelty

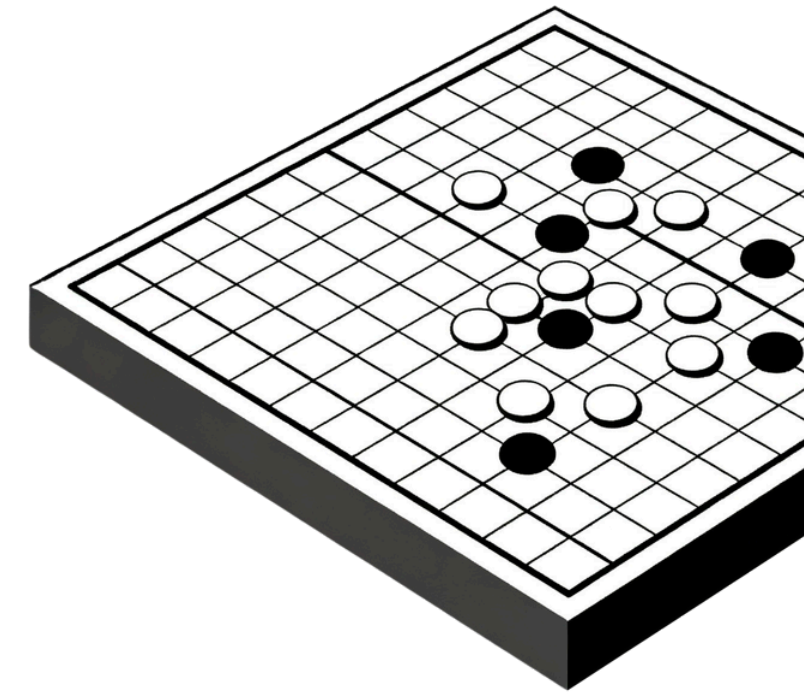
Prior Frameworks:

- AlphaGo (Silver et al., 2016),
- AlphaGo Zero (Silver et al., 2017),
- KataGo (Wu, 2019).

Theoretical Grounding:

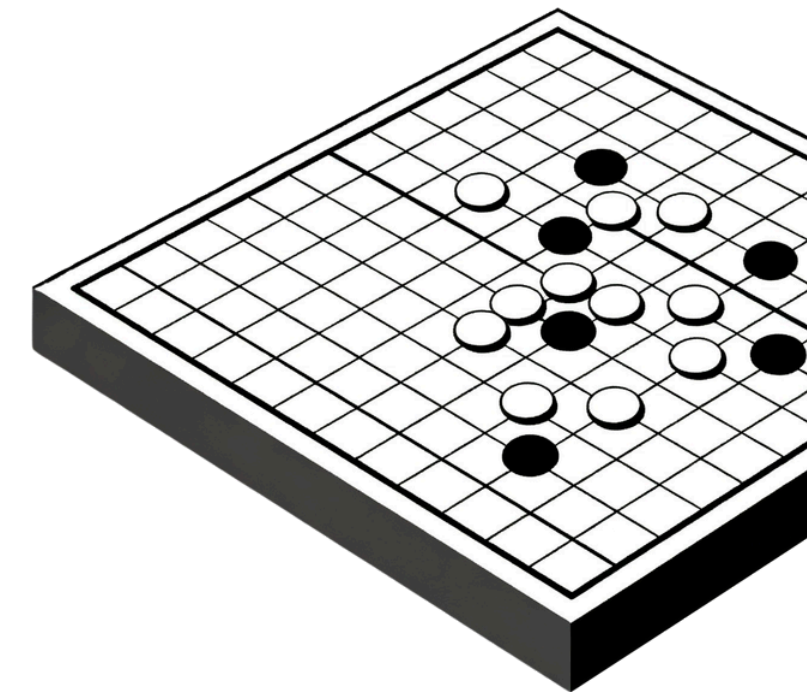
- Feature selection theory (Guyon & Elisseeff, 2003) and
- small-board Go benchmarks (van der Werf et al., 2003).

What ZetaGo Proves: Through formal statistical testing (ANOVA, McNemar), feature engineering and deep model capacity are shown to be strict substitutes, not complements.



Dataset

Dataset & Benchmark Baselines

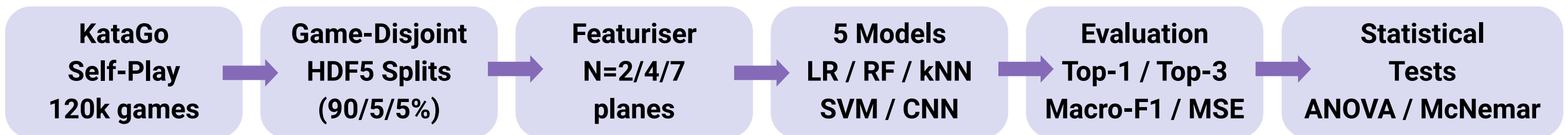
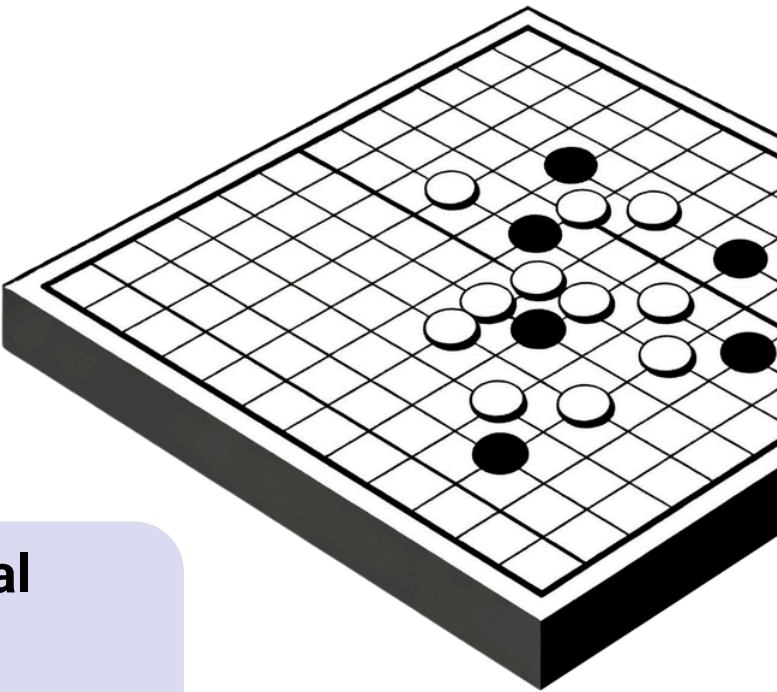


Source: 120,000 KataGo v1.16.5 self-play games on 7×7 board, komi 9.5 (area scoring), maxVisits=16. Resignation disabled, every game plays to a scored terminal.

Split	Games	Positions	Share
Train	107,969	3,310,156	90%
Validation	6,027	184,834	5%
Test	6,004	182,102	5%

Methodology

Experimental Design



Factor A → Feature Richness

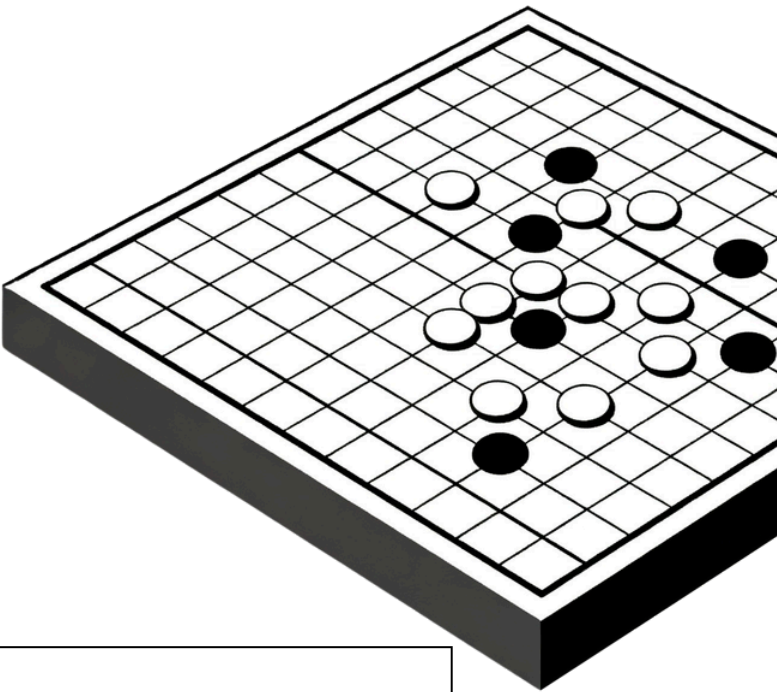
N = 2	Plane 1 → Current Player Stone Positions Plane 2 → Opponent Player Stone Positions
N = 4	Plane 3 → Normalised Group Liberties [0, 1] Plane 4 → Turn Indicator
N = 7	Plane 5 → Board State one turn ago Plane 6 → Board State two turn ago Plane 7 → Occupancy Change

Factor B → Data Volume

- 1k / 5k / 20k / full (108k) games sampled.
- 108× more games yields only 10.3× more unique positions — strong sublinear growth.
- 5 random seeds (42–46) per cell = 300 experiments.

Models & Architecture

Classical Baselines (sklearn)



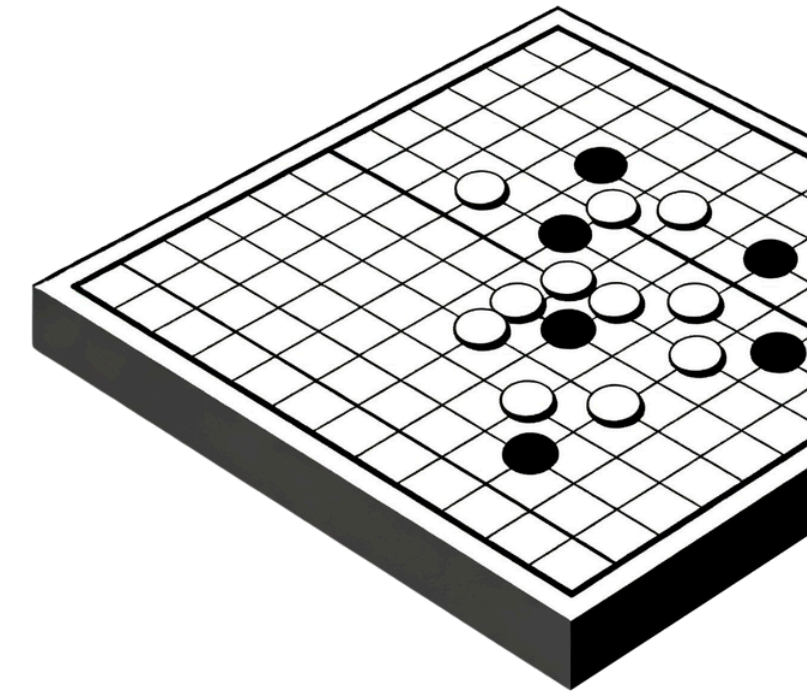
Model Class	Policy Head Config	Value Head Config	Execution Notes
Logistic Regression	Multinomial, LBFGS solver (max 500 iter)	Ridge Regressor	Flattened 1D input, capped at 80k rows
Random Forest	200 trees, max_features="sqrt"	Random Forest Regressor	Flattened 1D input, capped at 80k rows
k-NN	k=11, Euclidean distance	k-NN Regressor	Lazy learner – 0 train time, high inference cost
Linear SVM	Linear SVC with hinge loss	Linear SVR	Flattened 1D input, capped at 80k rows

Models & Architecture

CNN Champion (PyTorch)

- **Shared body:** $3 \times \text{Conv2d}$ (64 filters, 3×3 , padding=1) + ReLU — preserves 2D spatial geometry
- **Policy head:** 1×1 conv $\rightarrow$ flatten $\rightarrow$ Linear(50) $\rightarrow$ softmax
- **Value head:** 1×1 conv $\rightarrow$ flatten $\rightarrow$ Linear(64) $\rightarrow$ ReLU $\rightarrow$ Linear(1)
- **Training:** Adam ($\text{lr}=10^{-3}$), batch 512, 12 epochs, full 3.31M rows
- **Loss:** Weighted Policy+Value (CE + MSE/scale2)

The CNN is the only model that uses the native 2D board representation.



Results on Test set

Main Results : Sealed Test Set Performance

Baselines (Test Split)

- Majority-class: 9.17%
- Uniform-random-legal: 3.50%
- Side-to-move value: 86.44%

Generalisation

Every model's test accuracy lands within 0.002 of its validation accuracy; smaller than its own across-seed SD. No measurable generalisation gap

Model Class	Top-1 Acc	Top-3 Acc	Macro-F1
CNN (Champion)	0.9037 ± 0.0005%	98.72%	0.9158
Random Forest	0.8752 ± 0.0003%	96.00%	0.8826
k-NN (k=11)	0.8137 ± 0.0030%	93.64%	0.8110
Logistic Regression	0.7963 ± 0.0017%	91.96%	0.7952
SVM	0.7912 ± 0.0014%	91.57%	0.7878

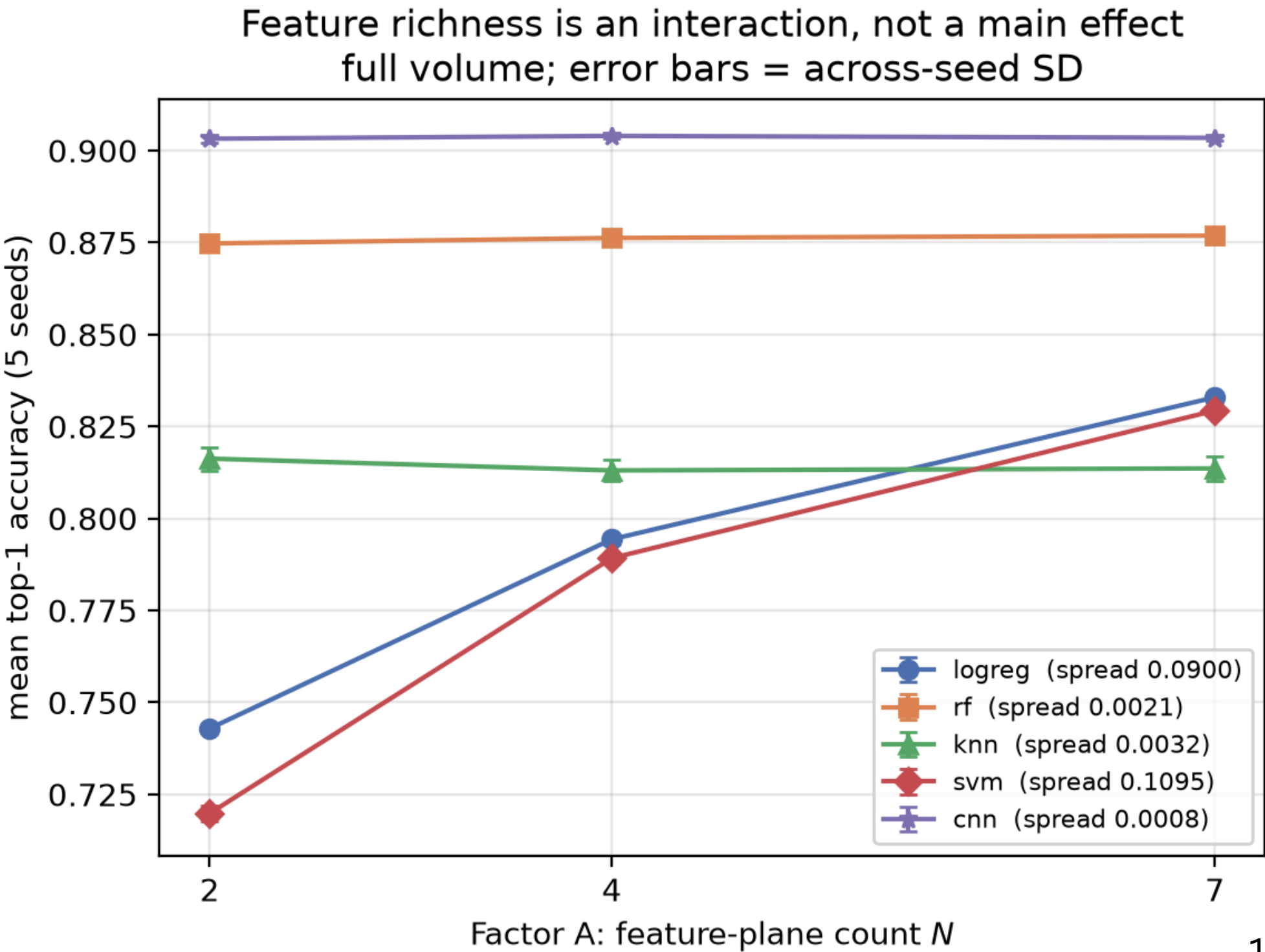
Discovery 1

Feature Engineering is Model-Dependent

Spread from N=2 to N=7 (full volume) vs. each model's seed SD:

Model	Spread	Seed SD	Ratio
SVM	0.1095	0.0016	70x
LogReg	0.0900	0.0012	77x
RF	0.0021	0.0005	4.7x
kNN	0.0032	0.0032	1.0x
CNN	0.0008	0.0008	1.0x

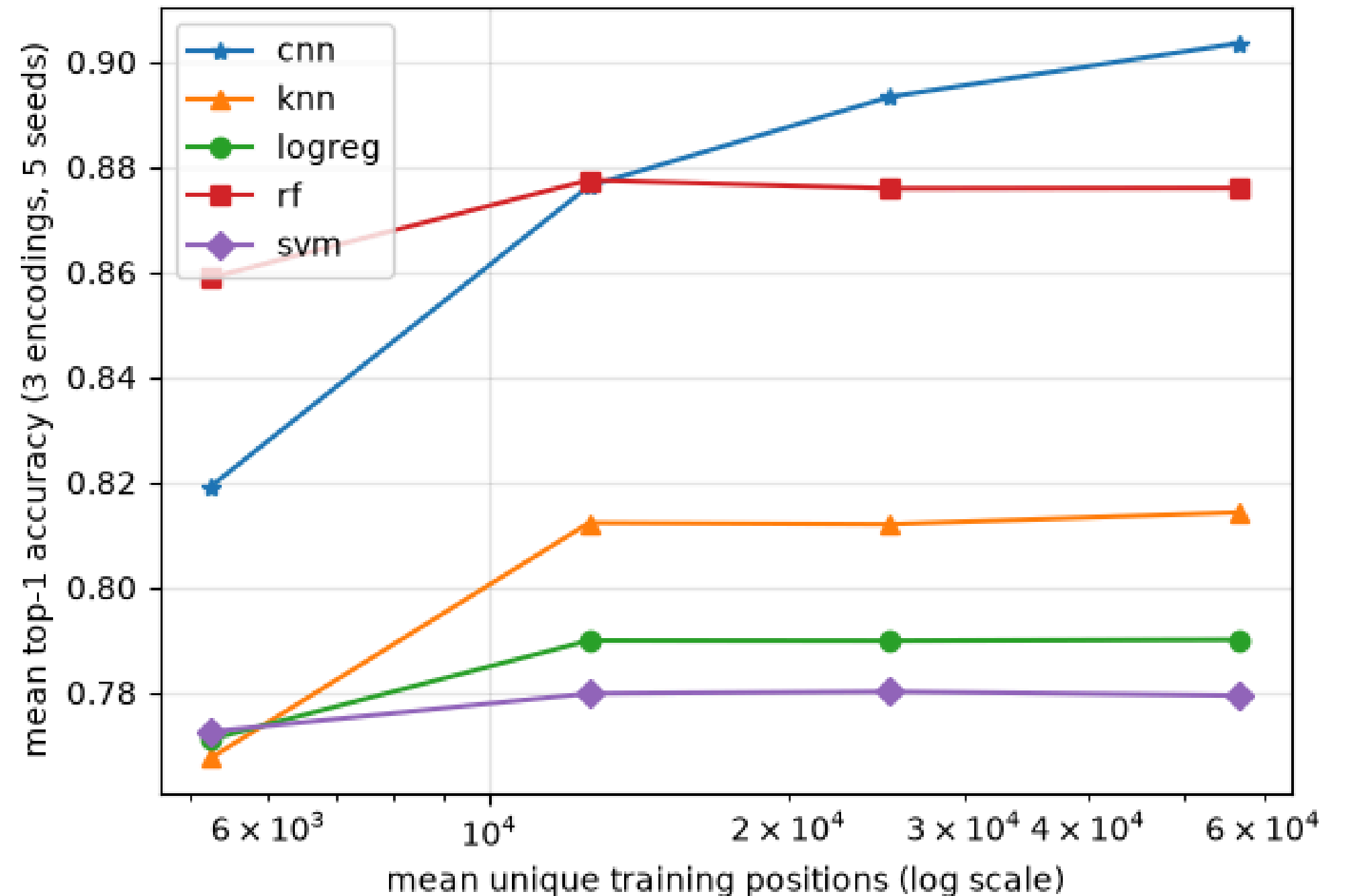
Linear models: massive +10% gain from hand-crafted features. CNN: gain is below seed noise and features are redundant.



Discovery 2

Data Volume Scaling & Crossovers

1. Random Forest dominates below 5k games leading to highest accuracy at 1k (0.859) thanks to bagging stability
2. CNN overtakes at ~13k unique positions; gap widens monotonically
3. Classical models hit a capacity ceiling: 5k→108k gives LR only +0.0002, SVM +0.0069
4. CNN still climbing at last data point crossover is a lower bound, not a plateau

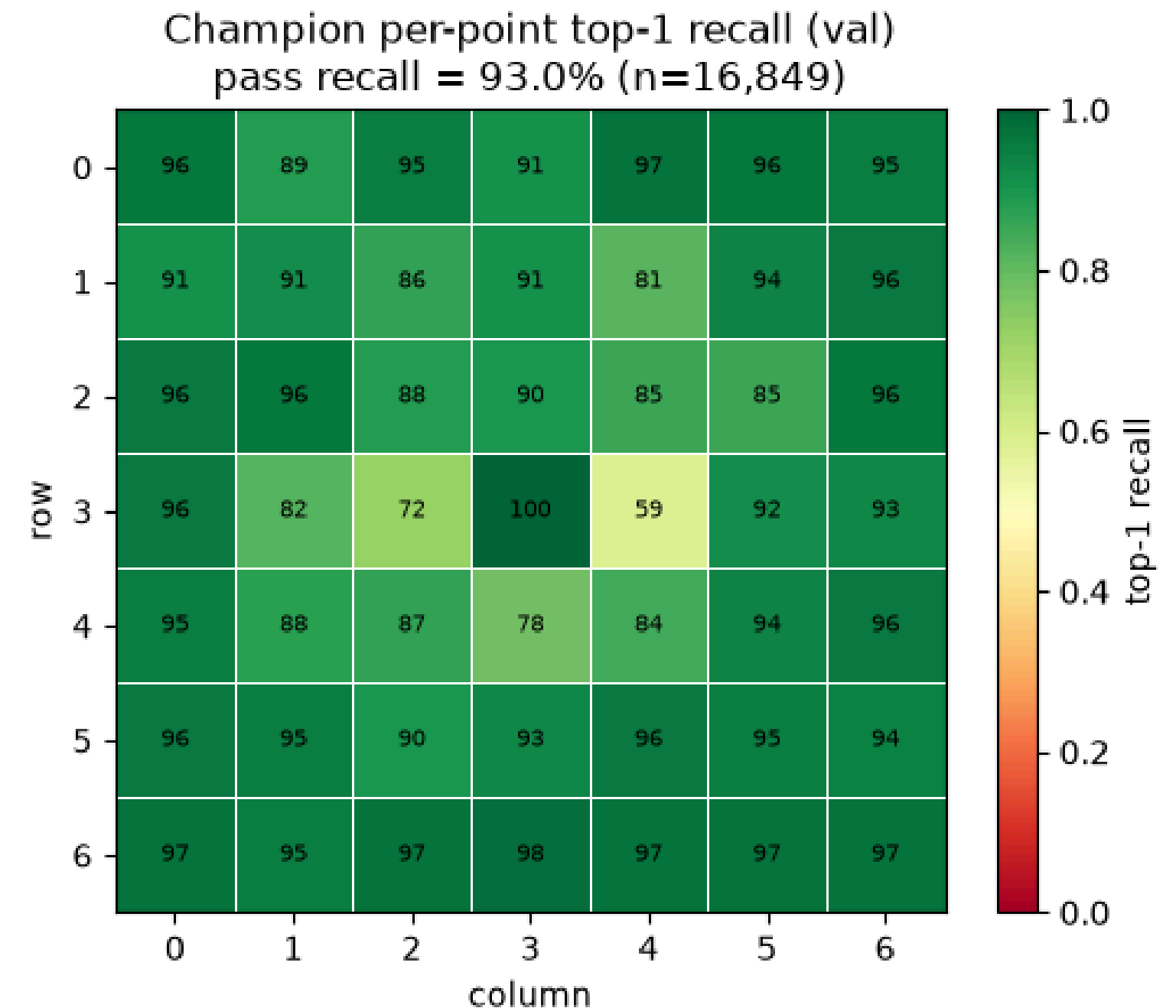


Discovery 3

Confusion & Spatial Recall Analysis

- Centre (3,3): 99.9% recall → nearly perfect
- Corner points (0,0), (6,0), etc. lowest recall (~75–80%)
- Orthogonal neighbours of centre: as low as 59.0% the hardest moves to predict
- pass: 93.0% recall; the model occasionally plays on instead of passing

Spatial recall heatmap for the CNN champion model on the 7x7 board. Centre point achieves 99.9% recall.



Evaluation

Value Prediction & Compute Budget

Model Class	Value Acc	MSE	vs BASE
Logistic Regression	0.712	119.2	-0.143
Linear SVM	0.780	133.6	-0.076
k-NN	0.924	80.2	+0.068
Random Forest	0.932	72.4	+0.076
CNN (Champion)	0.917	69.8	+0.0611

- LR and SVM lose to this trivial baseline where they learned nothing about position evaluation.
- RF has highest sign accuracy
- CNN has lowest MSE.

Model	Time	Share
SVM	6.31	53.8%
CNN	3.90	33.2%
LR	0.98	8.3%
RF	0.37	3.2%
k-NN	0.00	0.0%

SVM took 2× longer than CNN yet achieved far worse results, which shows compute vs. model capacity.

Key Insights & Discussion

Key Interpretations:

- Feature planes are substitutes for capacity. Handing a CNN pre-computed features is computationally redundant.
- Classical models are capacity-limited, not data-limited
- CNN wins MSE; RF wins sign accuracy. The RF draws sharper classification boundaries near margin zero.

An interesting finding:

The SVM is the most expensive model (6.3h) yet the worst performer in both compute time & capability.

Where and why the model fails:

- Flattened 1D input destroys spatial adjacency.
- Higher-dimensional feature planes hurt via the Curse of Dimensionality.
- Only the CNN is operationally viable for MCTS integration.

Limitations

Experimental Limitations

Hardware Constraints

12-hour GPU cap forced a 12-epoch limit on CNNs and an 80k row cap on classical algorithms.

Teacher Ceiling

Labels reflect KataGo's move distribution at maxVisits=16, evaluating imitation fidelity rather than perfect Go play.

Opening Determinism

Fixed opening moves resulted in only 17 distinct board positions by ply 4.



Conclusion & Future Work

Primary Conclusion:

Feature engineering and model capacity are strict substitutes. Hand-crafted features boost linear models (+11.0%) but add zero value to deep CNNs.

Future Research:

- Test scaling on larger board sizes (9x9, 19x19).
- Benchmark deeper architectures (ResNets, Vision Transformers).
- Introduce randomized opening play to eliminate deterministic opening turns.



References

- Mastering the game of Go with deep neural networks and tree search (Silver et al., 2016) - Heidelberg University
- Overview on DeepMind and Its AlphaGo Zero AI - Academia.edu
- Accelerating Self-Play Learning in Go
- AlphaGo - The Movie



Thank You!

Questions?